Probabilistic and fuzzy interpretations for relation-changing operators

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Outline.

- Relation changing structures.
- Relation-Changing operators.
- Fuzzy modal Logic.
- Fuzzy relation-changing logics.
- A Probabilistic Modal Logic.
- Probabilistic relation-changing logics.
- Conclusion and Future work.

Relation changing formalisms.

Graphs are fundamental structures in computer science and mathematics, which are able to model a relationship between entities.

While this relation is fixed in many cases, we can find several examples of models where this relation may change.

Dynamic features of graph can be used to model graph games, information flow and software verification, for instance.

Modal logics are a well-known class of logics used to reason about graph-like structures. In the literature we can find practical situations of these logics applied to relation-changing structures.

Relation changing formalisms.

Embed the relation-changes in the structure:

- Reactive graphs
- Switch graphs

Enrich modal logic with relation-changing operators:

- Sabotage logic
- Swap logic
- Bridge logic

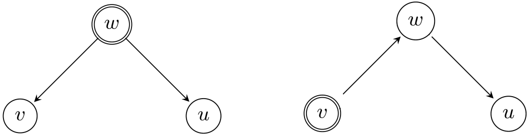
We focus in relation-changing modal logics and in the *local* version of these operators.

Relation changing formalisms.

Swap logic enriches usual modal logic with new operator $\blacklozenge$.

The semantics for the $\blacklozenge$ operator consider the same accessibility relation as the regular $\Diamond$, but the model changes.

Whenever an edge is crossed, the accessibility relation is modified by swapping the traversed edge.

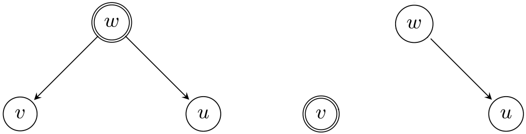


Relation changing structures.

Sabotage logic enriches usual modal logic with new operator $\blacklozenge$.

The semantics for the $\blacklozenge$ operator consider the same accessibility relation as the regular $\lozenge$, but the model changes.

Whenever an edge is crossed, the accessibility relation is modified by removing the traversed edge.

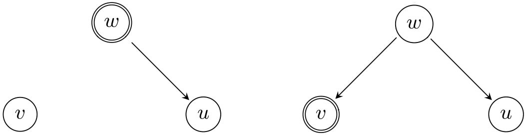


Relation changing structures.

Bridge also logic enriches usual modal logic with a new operator ⧫.

The semantics for the ⧫ operator consider the complement of the regular $\Diamond$ accessibility relation .

The operator connects two unrelated vertices, changing the accessibility relation by adding this new edge, which is also crossed.



Relation changing structures.

These logic are classic, in the sense that an edge either exists or it does not.

While this is useful for many cases, there are real-life systems which are not representable by this:

- 1. We may have fuzzy scenarios, where we are not sure about the existence of an edge;
- 2. Probabilistic cases where we can determine how likely it is to chose a certain path.

We propose to explore both cases.

Fuzzy Modal Logic.

Fuzzy formalism is a mean to overcome Boolean rigidity.

"The chemical reaction can only occur if is hot enough"

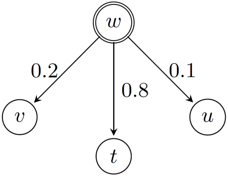
Instead of simply considering a binary set of possibilities (such as *true* and *false*), fuzzy contexts admit graded levels of certainty (values on [0,1]).

We consider fuzziness on the accessibility relation. Fuzziness on the valuation of propositions could also be considered but is not the focus of this work.

Fuzzy Modal Logic.

Definition (Fuzzy frame)

A *fuzzy frame* is a tuple $\mathfrak{F} = (W, R)$ where W is a set of states and $R : W^2 \to [0, 1]$ is a membership function indicating the membership degree of each edge.



Definition (Fuzzy Models)

A *fuzzy Kripke model* is a triple $\mathfrak{M} = (W, R, V)$ where (W, R) is a fuzzy frame and $V : \text{Prop} \to 2^W$ is the valuation function. Let $w \in W$, the pair $(\mathfrak{M}, w)$ is called a (fuzzy) *pointed model*.

Fuzzy Modal Logic.

Definition

A *t-norm* is a binary operation $*: [0,1] \times [0,1] \to [0,1]$ satisfying the following conditions:

(i) $*$ is commutative and associative;

(ii) $*$ is non-decreasing in both arguments;

(iii) $1 * x = x$ and $0 * x = 0$ for all $x \in [0,1]$.

With this, the fuzzy residuum $\Rightarrow: [0,1] \times [0,1] \to [0,1]$ is an operation satisfying:

$$x * z \leq y \text{ iff } z \leq (x \Rightarrow y)$$

and the fuzzy negation $\sim x$ is the abbreviation of $x \Rightarrow 0$.

Fuzzy Modal Logic.

Lemma

Let $*$ be a continuous t-norm. Then there is a unique operation $x \Rightarrow y$ satisfying, for all $x, y, z \in [0, 1]$,

$$x * z \leq y \text{ iff } z \leq (x \Rightarrow y)$$

namely: $x \Rightarrow y = \max\{z \mid x * z \leq y\}$

t-norm	Definition	Implication $a \Rightarrow b$	Negation $\sim a$
Łukasiewicz	$\max(0, a + b - 1)$	$\min(1, 1 - a + b)$	$1 - a$
Gödel	$\min(a, b)$	1 if $a \leq b$, else b	1 if $a = 0$, else 0
Product	$a \cdot b$	1 if $a \leq b$, else b/a	1 if $a = 0$, else 0

Fuzzy Modal Logic.

Definition

Let Prop be a set of atomic propositions, the set of modal formulas Form is given by the following grammar:

$$\varphi, \psi ::= p \mid \bot \mid \varphi \wedge \psi \mid \varphi \to \psi \mid \Diamond\varphi,$$

A *crisp* valuation function over the proposition of Prop (i.e., is either 0 or 1) is used, but a fuzzy version can also be defined straightforwardly.

Fuzzy Modal Logic.

Definition

Let $\mathfrak{M} = (W, R, V)$ be a fuzzy Kripke model, $w \in W$, and let T be a continuos t-norm. The semantics of a formula φ is given recursively by a function $[\![\cdot]\!]^w_{\mathfrak{M}} : \text{Form} \to [0,1]$ as:

$$
\begin{array}{lcl}
[\![p]\!]^w_{\mathfrak{M}} & = & \begin{cases} 1 & \text{if } w \in V(p) \\ 0 & \text{otherwise} \end{cases} \\
[\![\bot]\!]^w_{\mathfrak{M}} & = & 0 \\
[\![\varphi \wedge \psi]\!]^w_{\mathfrak{M}} & = & T([\![\varphi]\!]^w_{\mathfrak{M}}, [\![\psi]\!]^w_{\mathfrak{M}}) \\
[\![\varphi \to \psi]\!]^w_{\mathfrak{M}} & = & [\![\varphi]\!]^w_{\mathfrak{M}} \Rightarrow [\![\psi]\!]^w_{\mathfrak{M}} \\
[\![\Diamond\varphi]\!]^w_{\mathfrak{M}} & = & \sup\limits_{w' \in W} \left(T(R(w, w'), [\![\varphi]\!]^{w'}_{\mathfrak{M}}) \right).
\end{array}
$$

In the fuzzy case, the modal operator $\Box$ is not necessarily the dual of $\Diamond$. Thus, we are considering a $\Diamond$-fragment of the full modal fuzzy logic.

Fuzzy swap, sabotage and bridge logics.

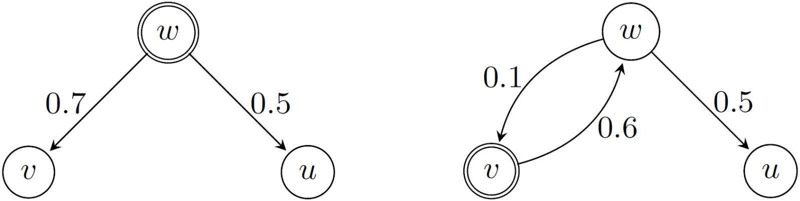
Fuzzy swap logic

Definition

We denote by $\mathrm{ML}(\langle \mathsf{sw} \rangle_x)$ the language of fuzzy swap logic. Let Prop be a set of atomic propositions, the set of formulas Form_{sw} is given by:

$$\varphi, \psi ::= p \mid \bot \mid \varphi \wedge \psi \mid \varphi \to \psi \mid \Diamond\varphi \mid \langle \mathsf{sw} \rangle_x \varphi$$

where $p \in \mathrm{Prop}$ and $x \in [0, 1]$.



$\langle \mathsf{sw} \rangle_{0.6}\varphi$

Let $\mathfrak{M} = (W, R, V)$ be a fuzzy Kripke model, $w \in W$, and let T be a t-norm. The semantics of a formula φ is given by a function $[\![\cdot]\!]^{w}_{\mathfrak{M}} : \mathrm{Form}_{sw} \to [0,1]$, extending the one of Fuzzy modal logic as:

$$[\![\langle \mathsf{sw} \rangle_x \varphi]\!]^{w}_{\mathfrak{M}} \quad = \quad \sup_{\{w' : R(w,w') \geq x\}} \Big(T(R(w,w'), [\![\varphi]\!]^{w'}_{\mathfrak{M}^{x}_{*(w',w)}}) \Big),$$

where, for fixed values of w' and x, $\mathfrak{M}^{x}_{*(w',w)} = (W, R^{x}_{*(w',w)}, V)$ with $R^{x}_{*(w',w)}$ defined in the following way:

$$R^{x}_{*(w',w)}(v,u) = \begin{cases} \min(R(v,u)+x, 1) & \text{if } (v,u) = (w',w) \text{ and } w \neq w' \\ R(v,u) - x & \text{if } (v,u) = (w,w') \text{ and } w \neq w' \\ R(v,u) & \text{otherwise.} \end{cases}$$

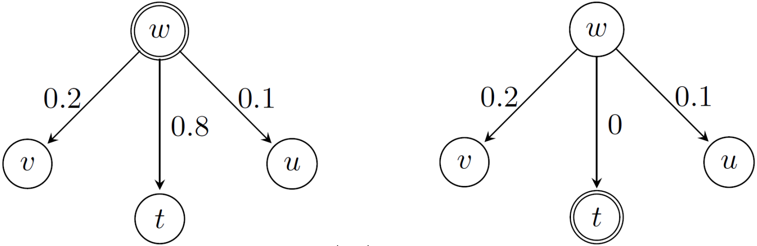
Fuzzy swap, sabotage and bridge logics.

Definition

We denote by $\mathrm{ML}(\langle \mathsf{sb} \rangle_x)$ the language of fuzzy sabotage logic. Let Prop be a set of atomic propositions, the set of formulas

$$\varphi, \psi ::= p \mid \bot \mid \varphi \wedge \psi \mid \varphi \to \psi \mid \Diamond\varphi \mid \langle \mathsf{sb} \rangle_x \varphi$$

where $p \in \mathrm{Prop}$ and $x \in [0, 1]$.



$\langle \mathsf{sb} \rangle_{0.8} \varphi$

The function $[\![\cdot]\!]^{w}_{\mathfrak{M}} : \mathrm{Form}_{sb} \to [0,1]$ extends the one Fuzzy modal logic as:

$$[\![\langle \mathsf{sb} \rangle_x \varphi]\!]^{w}_{\mathfrak{M}} \quad = \quad \sup_{\{w' : R(w,w') \geq x\}} \Big(T(R(w,w'), [\![\varphi]\!]^{w'}_{\mathfrak{M}^{x}_{-(w,w')}}) \Big),$$

where, for fixed values of w' and x, $\mathfrak{M}^{x}_{-(w,w')} = (W, R^{x}_{-(w,w')}, V)$ where $R^{x}_{-(w,w')}$ is defined in the following way:

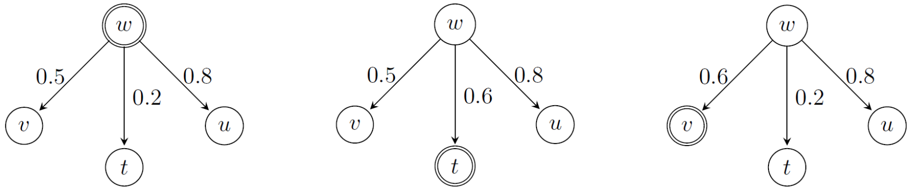
$$R^{x}_{-(w,w')}(v,v') = \begin{cases} R(v,v') - x & \text{if } (v,v') = (w,w') \\ R(v,v') & \text{otherwise.} \end{cases}$$

Definition

We denote by $\mathrm{ML}(\langle \mathrm{sb} \rangle_x)$ the language of fuzzy sabotage logic. Let Prop be a set of atomic propositions, the set of formulas

$$\varphi, \psi ::= p \mid \bot \mid \varphi \wedge \psi \mid \varphi \rightarrow \psi \mid \Diamond \varphi \mid \langle \mathrm{br} \rangle_x \varphi$$

where $p \in \mathrm{Prop}$ and $x \in [0,1]$.



$\langle \mathrm{br} \rangle_{0.6} \varphi$

The function $\llbracket \cdot \rrbracket^{w}_{\mathfrak{M}} : \mathrm{Form}_{br} \to [0,1]$ extends the one of Fuzzy modal logic as:

$$\llbracket \langle \mathrm{br} \rangle_x \varphi \rrbracket^{w}_{\mathfrak{M}} \quad = \quad \sup_{\{w' : R(w,w') < x\}} \big(T(x, \llbracket \varphi \rrbracket^{w'}_{\mathfrak{M}^x_{+(w,w')}}) \big),$$

where, for fixed values of w' and x, $\mathfrak{M}^x_{+(w,w')} = (W, R^x_{+(w,w')}, V)$ where $R^x_{(w,w')}$ is defined in the following way:

$$R^x_{+(w,w')}(v, v') = \begin{cases} x & \text{if } (v,v') = (w,w') \text{ and } R(w,w') < x \\ R(v,v') & \text{otherwise.} \end{cases}$$

Fuzzy swap, sabotage and bridge logics.

Proposition

For any φ, $\mathfrak{M}$ and w, we have that $[\![\Diamond\varphi]\!]^{w}_{\mathfrak{M}} = [\![\langle \mathsf{sw}\rangle_0\varphi]\!]^{w}_{\mathfrak{M}}$.

Proposition

For any φ, $\mathfrak{M}$ and w, we have that $[\![\Diamond\varphi]\!]^{w}_{\mathfrak{M}} = [\![\langle \mathsf{sb}\rangle_0\varphi]\!]^{w}_{\mathfrak{M}}$.

The same result does not hold for $\mathsf{ML}(\langle \mathsf{br}\rangle_x)$.

Bisimulation.

Definition (Swap/sabotage-bisimulations)

Let $\mathfrak{M} = (W, R, V)$ and $\mathfrak{M}' = (W', R', V')$ be two fuzzy Kripke models. A relation $Z \subseteq (W \times [0,1]^{W^2}) \times (W' \times [0,1]^{W'^2})$ is called a *swap* (respectively *sabotage*) *bisimulation* if and only if, for some $x \in [0,1]$, $(w, E)Z(w', E')$ implies:

atom: $w \in V(p)$ iff $w' \in V'(p)$, for all $p \in \mathrm{Prop}$;

swap/sab-zig: for any $v \in W$ such that $E(w, v) \geq x$, exists $v' \in W'$ s.t. $E'(w', v') \geq E(w, v)$ and $(v, E^x_{*(w,v)})Z(v', E'^x_{*(w',v')})$ (resp. $(v, E^x_{-(w,v)})Z(v', E'^x_{-(w',v')})$);

swap/sab-zag: for any $v' \in W'$ such that $E'(w', v') \geq x$, exists $v \in W$ s.t. $E(w, v) \geq E'(w', v')$ and $(v, E^x_{*(w,v)})Z(v', E'^x_{*(w',v')})$ (resp. $(v, E^x_{-(w,v)})Z(v', E'^x_{-(w',v')})$).

Bisimulation.

We say that $(\mathfrak{M}, w)$ and $(\mathfrak{M}', w')$ are x-swap (resp. x-sabotage) bisimilar iff there is a swap (resp. sabotage) bisimulation Z such that $(w, R)E(w', R')$.

They are swap (resp. sabotage) bisimilar, iff there is a swap (resp. sabotage) bisimulation Z such that $(w, R)E(w', R')$ for all $x \in [0, 1]$.

Theorem (Invariance under swap/sabotage bisimulation.)

Let $\mathfrak{M} = (W, R, V)$ *and* $\mathfrak{M}' = (W', R', V')$ *be two fuzzy Kripke models,* $w \in W, w' \in W'$.

- $[\![\varphi]\!]^{w}_{\mathfrak{M}} = [\![\varphi]\!]^{w'}_{\mathfrak{M}'}$ *for all formula* $\varphi \in \mathrm{Form}_{sw}$ *whenever* $(\mathfrak{M}, w)$ *and* $(\mathfrak{M}', w')$ *are swap-bisimilar.*
- $[\![\varphi]\!]^{w}_{\mathfrak{M}} = [\![\varphi]\!]^{w'}_{\mathfrak{M}'}$ *for all formula* $\varphi \in \mathrm{Form}_{sb}$ *whenever* $(\mathfrak{M}, w)$ *and* $(\mathfrak{M}', w')$ *are sabotage-bisimilar.*

Bisimulation.

Definition (Bridge-bisimulations)

Let $\mathfrak{M} = (W, R, V)$, $\mathfrak{M}' = (W', R', V')$ be two fuzzy Kripke models, $w \in W$ and $w' \in W'$.

A relation $Z \subseteq (W \times [0,1]^{W^2}) \times (W' \times [0,1]^{W'^2})$ is a *bridge bisimulation* if and only if, $(w, E)Z(w', E')$ implies:

- **atom:** $w \in V(p)$ iff $w' \in V'(p)$, for all $p \in \text{Prop}$;
- **bridge-zig:** for any $v \in W$ such that $E(w, v) < x$, exists $v' \in W'$ s.t. $x > E'(w', v')$ and $(v, E^x_{+(w,v)})Z(v', E'^x_{+(w',v')})$;
- **bridge-zag:** for any $v' \in W'$ such that $E'(w', v') < x$, exists $v \in W$ s.t. $x > E(w, v)$ and $(v, E^x_{+(w,v)})Z(v', E'^x_{+(w',v')})$.

Bisimulation.

We say that $(\mathfrak{M}, w)$ and $(\mathfrak{M}', w')$ are x-bridge bisimilar iff there is a bridge-bisimulation Z such that $(w, R)E(w', R')$.

They are bridge-bisimilar, iff there is a bridge-bisimulation Z such that $(w, R)E(w', R')$ for all $x \in [0, 1]$.

Theorem (Invariance under bridge-bisimulation.)

Let $\mathfrak{M} = (W, R, V)$ and $\mathfrak{M}' = (W', R', V')$ be two fuzzy Kripke models, $w \in W, w' \in W'$.

- $[\![\varphi]\!]^{w}_{\mathfrak{M}} = [\![\varphi]\!]^{w'}_{\mathfrak{M}'}$ *for all formula* $\varphi \in \mathrm{Form}_{br}$ *whenever* $(\mathfrak{M}, w)$ *and* $(\mathfrak{M}', w')$ *are bridge-bisimilar.*

Model checking.

model-checking problem under consideration.

Input: A finite fuzzy Kripke model $\mathfrak{M}$, a state w of $\mathfrak{M}$, and a formula φ.

Output: $[\![\varphi]\!]^{w}_{\mathfrak{M}}$.

Lemma

The model-checking problem for $\mathrm{ML}(\langle \mathrm{sw} \rangle_x)$, $\mathrm{ML}(\langle \mathrm{sb} \rangle_x)$ and $\mathrm{ML}(\langle \mathrm{br} \rangle_x)$ is in PSpace.

The result considers a standard labeling algorithm in the literature, and adds the reasoning for each one of the operator.

Theorem

The model-checking problem for the logics $\mathrm{ML}(\langle \mathrm{sw} \rangle_x)$, $\mathrm{ML}(\langle \mathrm{sb} \rangle_x)$ *and* $\mathrm{ML}(\langle \mathrm{br} \rangle_x)$ *is* PSpace-*complete.*

Probabilistic modal logic.

Probabilistic Kripke models, without relation-changing dynamics exist are already well studied.

For instance, probabilistic temporal logic can be used to reason about Markov chains (PRISM, Kwiatkowska)

Definition

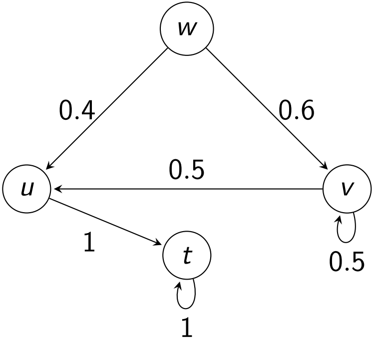
A *probabilistic Kripke model* is a tuple $\mathfrak{M} = (W, P, V)$ where:

- W is a finite set of states or worlds;
- $P : W \times W \to [0, 1]$ is the transition probability matrix where for all $w \in W$, $\sum_{w' \in W} P(w, w') = 1$;
- $V : \text{Prop} \to 2^W$ is a function that assigns to each $p \in \text{Prop}$ the set of states in which p holds.

Probabilistic modal logic.

Although probabilistic formalism admits values between 0 and 1 (thus being redeemed as a fuzzy formalisms particular case), these are usually study *per se due to its relevance in applied contexts.*

Probabilistic modal logic.



For each state, the probabilities of every edges leaving sum to 1.

Probabilistic modal logic.

Given a set Prop of proposition, the set of formulas is defined recursively as:

$$\varphi, \psi ::= p \mid \neg\varphi \mid \varphi \vee \psi \mid \Diamond_\rho \varphi,$$

where $\rho \in [0, 1]$.

Definition (Satisfaction)

We define the satisfaction relation $\models$ in a model $\mathfrak{M} = (W, P, V)$ and a state $w \in W$ recursively as follows:

$$\begin{array}{lcl}
\mathfrak{M}, w \models p & \iff & w \in V(p) \\
\mathfrak{M}, w \models \neg\varphi & \iff & \mathfrak{M}, w \not\models \varphi \\
\mathfrak{M}, w \models \varphi \vee \psi & \iff & \mathfrak{M}, w \models \varphi \text{ or } \mathfrak{M}, w \models \psi \\
\mathfrak{M}, w \models \Diamond_\rho \varphi & \iff & \displaystyle\sum_{\{v \mid \mathfrak{M}, v \models \varphi\}} P(w, v) \geq \rho.
\end{array}$$

Probabilistic modal logic.

Let $\mathfrak{M} = (W, P, V)$ and $\mathfrak{M}' = (W', P', V')$ be two probabilistic Kripke models and $E \subseteq W \times W'$. We say that a nonempty relation E is a *bisimulation from to '* if for every $w \in W$ and $w' \in W'$ such that wEw', we have

atoms: for any $p \in \mathrm{Prop}$, $w \in V(p)$ iff $w' \in V'(p)$,

zigzag$_{\text{prob}}$: for any $U \subseteq W$, $\sum_{u \in U} P(w, u) \leq \sum_{u' \in E[U]} P'(w', u')$

Theorem

Let $\mathfrak{M} = (W, P, V)$ and $\mathfrak{M}' = (W', P', V')$ be two probabilistic Kripke models and $E \subseteq W \times W'$ a bisimulation from $\mathfrak{M}$ to $\mathfrak{M}'$. Then for any $(w, w') \in E$ and any formula φ we have

$$\mathfrak{M}, w \models \varphi \text{ iff } \mathfrak{M}', w' \models \varphi$$

Probabilistic modal logic.

While the model is probabilistic, we consider a logic whose semantics only considers two truth values.

Probabilistic Kripke models can be seen as a particular case of Fuzzy graphs. However, methods for studying both cases are different, due to the specificity of probabilistic framework.

When relation-changing fuzzy operators are applied to a probabilistic Kripke model, they do not generally preserve the required probabilistic properties.

We propose proper probabilistic operators and study them, by generalizing the classic sabotage and bridge operators.

Probabilistic relation-changing logics.

The language we use is given by the following BNF:

$$\varphi, \psi ::= p \mid \neg\varphi \mid \varphi \vee \psi \mid \Diamond_\rho \varphi \mid \blacklozenge_\sigma \varphi,$$

where $p \in \text{Prop}$, $\rho \in [0, 1]$ and $\sigma \in]0, 1[$.

The semantics for $\blacklozenge$ operator are presented for both probabilistic sabotage and probabilistic bridge contexts.

We exclude $\sigma = 0$ and $\sigma = 1$ due to arithmetic and functional restrictions to operator semantics.

Probabilistic relation-changing logics.

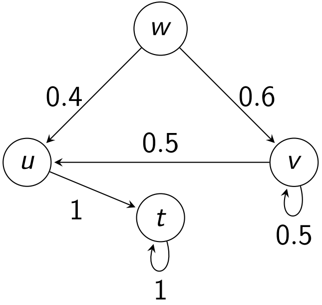
Given a probabilistic Kripke model $\mathfrak{M}$, the semantics for probabilistic sabotage logic are the same as probabilistic modal logic with the addition of semantics for the following operator:

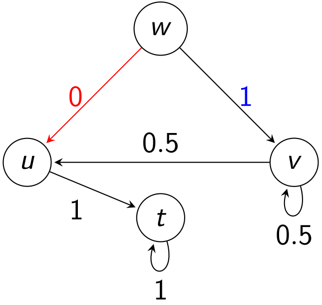
$\mathfrak{M}, w \models \blacklozenge_\rho \varphi \iff$ exists $v \in W$ s.t. $\rho \leq \ P(w, v) \neq 1$,
and $\mathfrak{M}_\rho^{-(w,v)}, v \models \varphi$,

where $\mathfrak{M}_\rho^{-(w,v)} = (W, P_\rho^{-(w,v)}, V)$, with:

- $P_\rho^{-(w,v)}(w, v) = P(w, v) - \rho$;
- $P_\rho^{-(w,v)}(w, t) = P(w, t) + \dfrac{\rho.P(w, t)}{1 - P(w, v)}$, if $t \neq v$;
- $P_\rho^{-(w,v)}(s, t) = P(s, t)$, if $s \neq w$.

Probabilistic relation-changing logics.





$\mathfrak{M}, w \models \blacklozenge_{0.4}\varphi$

$\mathfrak{M}_{0.4}^{-(w,u)}, u \models \varphi$

Probabilistic relation-changing logics.

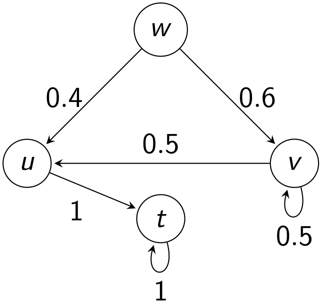
Given a probabilistic Kripke model $\mathfrak{M}$, the semantics for probabilistic bridge logic are the same as probabilistic modal logic with the addition of semantics for the following operator:

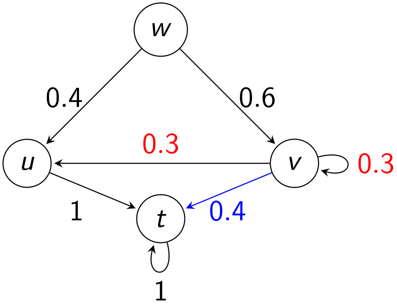
$\mathfrak{M}, w \models \blacklozenge_\rho \varphi \iff$ exists $v \in W$ s.t. $P(w,v) + \rho \leq 1$ and $\mathfrak{M}_\rho^{+(w,v)}, v \models \varphi$,

here $\mathfrak{M}_\rho^{+(w,v)} = (W, P_\rho^{+(w,v)}, V)$, with:

- $P_\rho^{+(w,v)}(w,v) = P(w,v) + \rho$;
- $P_\rho^{+(w,v)}(w,t) = P(w,t) - \dfrac{\rho . P(w,t)}{1 - P(w,v)}$, if $t \neq v$;
- $P_\rho^{+(w,v)}(s,t) = P(s,t)$, if $s \neq w$.

Probabilistic relation-changing logics.





$\mathfrak{M}, v \models \blacklozenge_{0.4}\varphi$

$\mathfrak{M}^{+(v,t)}_{0.4}, t \models \varphi$

Sabotage bisimulation.

Let $\mathfrak{M} = (W, P, V)$ and $\mathfrak{M}' = (W', P', V')$ be two probabilistic Kripke models. A ρ-sabotage *bisimulation from* $\mathfrak{M}$ *to* $\mathfrak{M}'$ is a non-empty relation $E \subseteq \big(W \times Prob(W)\big) \times \big(W' \times Prob(W')\big)$ when the following holds:

atom: $w \in V(p)$ iff $w' \in V'(p)$, for all $p \in \mathrm{Prop}$;

zigzag$_{\text{prob}}$: $\forall\, U \subseteq W, \sum_{u \in U} P(w,u) \leq \sum_{(u',P') \,\in\, \bigcup_{u \in U} E[(u,P)]} P'(w',u')$;

r−zig$_{\text{prob}}$: $\forall\, u \in W$ s.t. $1 \neq P(w,u) > \rho$, exists $u' \in W'$ such that:

1. $P(w,u) \leq P'(w',u') < 1$ and
2. $\big(u, P_{\rho}^{-(w,u)}\big) E \big(u', P_{\rho}'^{-(w',u')}\big)$;

r−zag$_{\text{prob}}$: $\forall\, u' \in W'$ s.t. $1 \neq P'(w',u') > \rho$, exists $u \in W$ such that:

1. $P'(w',u') \leq P(w,u) < 1$ and
2. $\big(u, P_{\rho}^{-(w,u)}\big) E \big(u', P_{\rho}'^{-(w',u')}\big)$.

Bridge bisimulation.

A ρ-bridge *bisimulation from* $\mathfrak{M}$ *to* $\mathfrak{M}'$ is a non-empty relation $E \subseteq \big(W \times Prob(W)\big) \times \big(W' \times Prob(W')\big)$ when the following holds:

atom & zigzag$_{\mathbf{prob}}$: (same as before)

r−zig$_{\mathbf{prob}}$: for any $u \in W$ s.t. $P(w,u) + \rho < 1$, exists $u' \in W'$ such that:

1. $P(w,u) \geq P'(w',u')$, and
2. $\big(u, P_\rho^{+(w,u)}\big) E \big(u', P_\rho'^{+(w',u')}\big)$;

r−zag$_{\mathbf{prob}}$: for any $u' \in W'$ s.t. $P'(w',u') + \rho < 1$, exists $u \in W$ such that:

1. $P'(w',u') \geq P(w,u)$ and
2. $\big(u, P_\rho^{+(w,u)}\big) E \big(u', P_\rho'^{+(w',u')}\big)$.

and, for bridge bisimulation only, $\forall\ w \in W, w' \in W'$, $E[\{(w,P)\}] \neq \emptyset \neq E^{-1}[\{(w',P')\}]$.

Bisimulation.

We say that E is a sabotage bisimulation if it is a ρ-sabotage bisimulation for every $\rho \in [0, 1]$.

If E is a bisimulation and $(w, P)E(w', P')$, we say that w and w' are two sabotage bisimilar states.

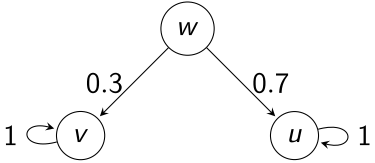
We say that E is a bridge bisimulation if it is a ρ-bridge bisimulation for every $\rho \in [0, 1]$.

If E is a bisimulation and $(w, P)E(w', P')$, we say that w and w' are two bridge bisimilar states.

Bisimulation.

Probabilistic sabotage and bridge logics are more expressive, being able to differentiate more probabilistic Kripke models.

Example of bisimilar models ($\mathfrak{M}$ and $\mathfrak{M}$') that are not sabotage-bisimilar not bridge-bisimilar:



$\mathfrak{M}$

$\mathfrak{M}'$

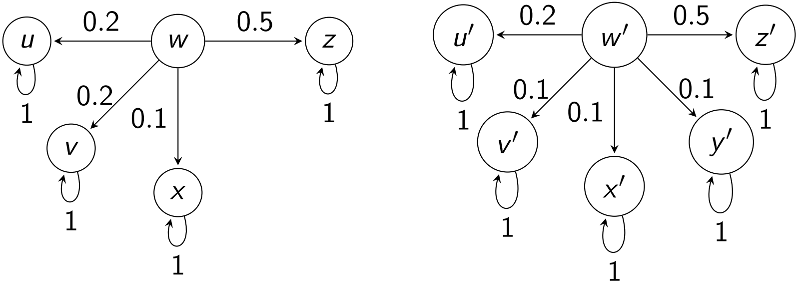
Bisimulation.

Theorem (Invariane)

Let $\mathfrak{M} = (W, P, V)$ and $\mathfrak{M}' = (W', P', V')$ be two probabilistic Kripke models and let $E \subseteq \big((W \times Prob(W)) \times \big(W' \times Prob(W')\big)$ be a sabotage (resp. bridge) bisimulation.

Then, $(w, P)E(w', P')$ implies $\mathfrak{M}, w \models \varphi$ iff $\mathfrak{M}, w' \models \varphi$, for every formula φ, under sabotage (resp. bridge) semantics.

Bisimulation.



$$E = \{((w,P),(w',P')),\ ((z,P),(z',P'))\}$$
$$\cup\ \{(t,P)\times(t',P') :\ t\in\{u,v,x\} \text{ and } t'\in\{u',v',x',y'\}\}$$
$$\cup\ \{(z,P_\rho^{-(w,z)})\times(z',P_\rho'^{-(w',z')}) :\ \rho\in]0,0.5]\}$$
$$\cup\ \{(t,P_\rho^{-(w,t)})\times(u',P_\rho'^{-(w',u')}) :\ t\in\{u,v\} \text{ and } \rho\in]0,0.2]\}$$
$$\cup\ \{(x,P_\rho^{-(w,x)})\times(t',P') :\ t'\in\{v',x',y'\} \text{ and } \rho\in]0,0.1]\}$$

is a sabotage-bisimulation.

Conclusion and future work.

We take the initial steps in studying relation changes for fuzzy sabotage, swap, and bridge operators. We define bisimulations and show the modal invariance; and prove that model-checking problem is PSpace-complete.

Following the work on fuzzy sabotage, swap, and bridge operators, we proposed sabotage and bridge operators for probabilistic framework. We define bisimulation and show the modal invariance.

We study the complexity of model checking for fuzzy relation-changing contexts.

We also would like to study the decidability status/complexity of the model checking problem and propose an automatic model-checker that integrates probabilistic sabotage and bridge operators. We would like to study tools such as PRISM that was adapted to relation-changing contexts.